

Protect the Decoder: Representation-Preserving Adaptation and Failure Detection for Far-OOD MedSAM

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Abstract. Parameter-efficient adaptation of MedSAM to one imaging domain can leave it worse than zero-shot on distant domains, a collapse that earlier work linked to drift of the mask decoder’s representations from the base model. We intervene on that mechanism with a centred kernel alignment (CKA) loss that pulls the adapted decoder’s activations on a 32-image out-of-distribution probe back toward the base decoder during LoRA training on dermoscopy. Over three seeds it raises tight-box Dice by 0.05 on breast ultrasound and 0.10 on mammography at no in-domain cost; anchoring the encoder alone lowers far-OOD Dice in every seed, and anchoring both halves gives the largest gain (0.18 Dice on mammography). Across adaptation budgets from 50 to 2595 images, LoRA falls below zero-shot far-OOD at 250 images, exactly where its far-OOD decoder drift triples, whereas decoder-only and encoder-only adaptation never collapse and never drift far-OOD: the failure requires joint, unanchored adaptation. A held-out mammography set reproduces the ranking. Finally, per-image decoder drift from the base model ranks LoRA’s far-OOD failures without ground truth, with AUROC 0.96 on ultrasound and 0.80 on mammography over three seeds (0.69 on the held-out set); flagging the 20 percent of ultrasound images with the highest drift captures 87 percent of failures, while the decoder’s own IoU estimate is at chance and responds to poor boxes instead. Far-OOD robustness of adapted MedSAM is decided at the decoder, where it can be protected and, for LoRA, monitored.

Keywords: Medical image segmentation · Parameter-efficient adaptation · OOD robustness · Failure detection

1 Introduction

MedSAM [13] adapts the Segment Anything Model [8] to medical images and is routinely fine-tuned again for a target modality with LoRA [6] or visual prompt tuning [7]. A prior comparison on a four-dataset drift ladder showed that the cheapest methods are also the most fragile: LoRA trained on dermoscopy gains in-domain accuracy and then loses 0.18 Dice relative to zero-shot MedSAM on mammography, while decoder-only and full fine-tuning keep their far-OOD accuracy [2]. The same study found that the loss correlates with how far the mask

decoder’s representations move from the base model, measured by centred kernel alignment (CKA) [9], and that encoder-side movement does not predict it. This paper asks whether that correlation is a mechanism that can be acted on, and what it needs to occur. That fine-tuning distorts pretrained features and hurts out of distribution is established [10,16]; the hypothesis tested here is narrower: for a promptable segmenter, far-OOD loss depends on where in the segmentation pathway the representations move, not on how much they move. First, does regularising the decoder toward the base model during adaptation reduce far-OOD collapse, and is the decoder specifically the locus? We add a CKA loss on a 32-image out-of-distribution probe and compare hooks on the decoder, the encoder, and both. Second, does the collapse scale with the adaptation budget? We train three adapters on nested subsets of 50, 250 and 1000 images at a fixed step budget. Third, can far-OOD failures be detected at inference without ground truth? We compare the IoU estimate SAM’s decoder already produces with the drift of the decoder’s upscaled mask embedding from the base model on the same image and prompt.

Our contributions are:

1. CKA-regularised LoRA with an OOD-only probe, three seeds, five jitter levels and a three-seed hook-placement ablation: encoder-only anchoring hurts, decoder anchoring helps, both is best;
2. a budget study with drift curves: LoRA’s far-OOD loss appears at 250 training images with a jump in far-OOD decoder drift, while decoder-only and encoder-only adaptation never collapse or drift, so the failure requires joint, unanchored adaptation;
3. a label-free per-image failure signal from decoder drift, assessed by threshold sensitivity, risk-coverage and a held-out domain (AUROC 0.96, 0.80 and 0.69 for LoRA over three seeds), and the finding that the decoder’s IoU head responds to poor prompts, not to modality shift.

2 Related Work

Adapting SAM to medical images. MedSAM fine-tunes SAM’s encoder and decoder on over a million image-mask pairs with box prompts [13]; SAM-Med2D adds encoder adapters [4]. Downstream users adapt these models again to one modality with LoRA [6], prompt tokens [7], or by unfreezing the decoder. The prior study we build on compared seven such strategies on a dermoscopy to ultrasound to mammography ladder and found that LoRA and prompt tuning collapse at the far end while decoder-only and full fine-tuning do not [2]. We take its testbed and diagnosis as given and study interventions.

Feature distortion under fine-tuning. Fine-tuning a good pretrained extractor can distort its features and underperform linear probing out of distribution, and in a linear model the distortion persists along the whole fine-tuning trajectory [10]; the same mechanism has been analysed for LoRA through the neural tangent kernel [17]. The remedies form one family: choose which layers to tune

by the type of shift [12], interpolate fine-tuned and zero-shot weights [19], or penalise the change of pretrained features directly, with a feature-protection loss for CLIP [16] or a penultimate-layer distance that preserves OOD detection after fine-tuning [21]; for MedSAM, adapting image embeddings at test time avoids parameter updates altogether [3]. Preserving pretrained representations is therefore not new. Those results concern one extractor read out by a linear head and preserve the total change of its features. A promptable segmenter has a second network between encoder and output, and our question is not whether representations move but where: the loss and recovery of robustness sit in the decoder, and encoder movement by itself is benign. We use linear CKA [9] as a differentiable anchor to the model’s own initialisation on a fixed probe batch.

Failure detection for segmentation. Estimating segmentation quality without ground truth ranges from reverse classification accuracy [18] to recent benchmarks of confidence-based failure detectors for medical segmentation [20]. SAM’s decoder emits a scalar IoU estimate for each mask [8], but its behaviour after adaptation has not been examined. We evaluate it against a detector derived from the mechanism studied here.

3 Method

f_θ denotes MedSAM with image encoder E_θ and mask decoder D_θ , θ_0 its released weights and θ the weights during adaptation; prompts are single boxes.

3.1 CKA-regularised adaptation

For activation matrices $X \in \mathbb{R}^{n \times p}$ and $Y \in \mathbb{R}^{n \times q}$ with centred columns, linear CKA is

$$\text{CKA}(X, Y) = \frac{\|Y^\top X\|_F^2}{\|X^\top X\|_F \|Y^\top Y\|_F} \in [0, 1] \quad (1)$$

[9]. Let P be a fixed probe batch of n images with boxes and $A_\ell(\theta; P)$ the activation captured by a forward hook at layer ℓ , one row per probe image. With $A_\ell(\theta_0; P)$ cached once, the training objective is

$$\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda \sum_{\ell \in H} w_\ell (1 - \text{CKA}(A_\ell(\theta_0; P), A_\ell(\theta; P))), \quad (2)$$

where H is the set of hooked layers, w_ℓ per-layer weights and $\mathcal{L}_{\text{task}}$ the equally weighted sum of binary cross-entropy and soft Dice. The probe forward runs every fourth step with λ multiplied by four (same expected gradient, a quarter of the cost), in single precision. The probe is OOD-only: 16 breast-ultrasound and 16 mammography images drawn once from the training splits of those datasets and absent from every evaluation set, each with a synthetic centred box covering half the image; an in-domain anchor would compete with the task loss, which already governs in-domain behaviour. We use $\lambda = 10$ (preliminary sweep over $\{0.1, 1, 10\}$).

3.2 Hook placement

Decoder hooks capture the output of the decoder’s upscaling block ($32 \times 256 \times 256$), the IoU prediction head and the mask logits, with $w_\ell = 0.8, 1.5$ and 1.0 , following the strength of the per-layer correlation reported in prior work [2]. *Encoder* hooks capture the neck output ($256 \times 64 \times 64$) and the outputs of the last two transformer blocks ($64 \times 64 \times 768$), each with $w_\ell = 1$. *Both* is the union. Every configuration backpropagates through the encoder, where the LoRA weights live; the positions differ only in which representation is anchored.

3.3 Adaptation-budget protocol

One fixed permutation of the training set (seed 0) supplies its first 50, 250 and 1000 images, so smaller budgets are prefixes of larger ones and every method sees identical images. Each run performs 6000 optimiser steps at batch size one under a cosine schedule, validating every 500 steps and keeping the best checkpoint; the full-data point is each method’s seed-0 checkpoint. Prompts are tight.

3.4 Per-image failure signals

For a test image x with box b , two label-free signals are available. The first is the decoder’s IoU estimate $\hat{u}(x, b)$, a scalar SAM predicts alongside each mask [8] and that adaptation updates with the rest of the decoder. The second is the decoder drift

$$d(x, b) = 1 - \text{CKA}(U_{\theta_0}(x, b), U_\theta(x, b)), \quad (3)$$

where $U(x, b) \in \mathbb{R}^{65536 \times 32}$ is the decoder’s upscaled mask embedding for that image and prompt, with the 256×256 positions as samples and the 32 channels as features. The same quantity on the encoder’s neck output (4096×256) gives an encoder drift. Computing them costs one extra base decoder forward per image, plus a base encoder forward when the encoder was adapted. A failure is a prediction with Dice below 0.5 (0.3 to 0.7 in a sensitivity check); both signals are scored by AUROC and AUPRC (high drift or low $\hat{u}$ positive) and by the recall and residual failure rate obtained when the top fraction of images by signal is flagged for review, and $\hat{u}$ is compared with the true IoU by a ten-bin expected calibration error.

4 Experimental Setup

Drift ladder. All adaptation uses the ISIC 2018 dermoscopy training set [5] (2594 training, 100 validation, 1000 test images). Evaluation adds PH2 [14] (200 dermoscopy images from another clinic, near-OOD), BUSI [1] (647 benign and malignant breast-ultrasound images, far-OOD) the 362 CBIS-DDSM test mammograms [11] with mass and calcification masks merged per image (far-OOD), and, as a held-out check used once, the 269 annotated mammograms of DMID [15] with their lesion masks (far-OOD). Images are resized to 1024×1024 ; box prompts come from the ground-truth mask, tight at evaluation unless stated.

Table 1. LoRA with and without the decoder CKA loss (20px training, tight-box evaluation), mean $\pm$ std over three seeds; Δ HD95 in pixels; p : paired Wilcoxon over per-image Dice at 20px evaluation jitter, Holm-corrected.

Dataset	no-CKA Dice	CKA Dice	Δ Dice	Δ HD95 (px)	p (Holm)
ISIC (ID)	0.949 ± 0.002	0.952 ± 0.002	+0.003	-0.3	<0.001
PH2 (near-OOD)	0.951 ± 0.001	0.954 ± 0.002	+0.003	-0.4	<0.001
BUSI (far-OOD)	0.774 ± 0.032	0.826 ± 0.008	+0.051	-96.1	<0.001
CBIS-DDSM (far-OOD)	0.437 ± 0.095	0.541 ± 0.095	+0.104	-152.4	<0.001

Methods. The seven strategies are those of the prior study [2]: zero-shot MedSAM; decoder-only fine-tuning; VPT-shallow and VPT-deep (ten prompt vectors at the encoder input or at every block, decoder trainable); LoRA with rank 8 on the fused query-key-value projection of all twelve encoder blocks, decoder trainable; encoder-only LoRA with rank 28 on every encoder linear layer, decoder frozen; and full fine-tuning. All use AdamW with a cosine schedule and mixed precision, with learning rates 5×10^{-4} (LoRA, prompts), 10^{-4} (decoder-only) and 10^{-5} (full). Regularised runs are LoRA plus Eq. (2). Training regime pm20 jitters each box corner uniformly within ± 20 pixels, pm0 uses tight boxes, and rand100 draws the jitter magnitude uniformly from $[0, 100]$ per step. A run takes six epochs on one 48 GB GPU.

Evaluation. We report Dice, IoU and HD95 (pixels on the 1024 grid). Prompt robustness expands each side of the tight box outward by an amount drawn uniformly from $[0, p]$ pixels, $p \in \{20, 50, 100, 200\}$, with one draw per image seeded on its identifier so all models see identical boxes. Multi-seed comparisons use seeds 0 to 2 and a paired Wilcoxon signed-rank test over per-image Dice, pairs matched on seed and image, Holm-corrected over the four datasets.

5 Results: CKA Regularisation

Multi-seed effect. Table 1 gives the main comparison. The decoder CKA loss leaves in-domain and near-OOD Dice within 0.003 and raises far-OOD Dice by 0.051 on BUSI and 0.104 on CBIS-DDSM, with HD95 falling by 96 and 152 pixels: the regulariser removes the boundary failures of unregularised LoRA rather than small overlap losses. The BUSI gain is stable across seeds (std 0.008 against 0.032 without the loss); on CBIS-DDSM the seed spread is large for both models (std 0.095) but the gain holds in every seed, and the paired test is significant on all four datasets. With tight-box training the loss gives smaller same-signed gains (0.033, 0.027; seed 0); with rand100 training it hurts (0.059 and 0.032 lower), so the two regularisers compound at mild jitter and conflict under heavy jitter.

Prompt robustness. In Fig. 1 the regularised pm20 model is the best of the six on both far-OOD sets at every expansion, with an advantage that narrows

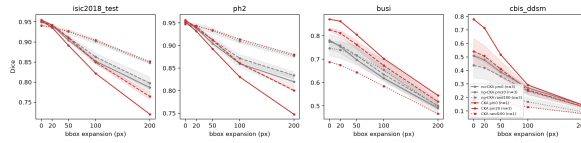


Fig. 1. Dice against outward box expansion for LoRA with (red) and without (grey) the decoder CKA loss under three training regimes: tight (solid), 20 px jitter (dashed), random jitter up to 100 px (dotted). Bands: one std over three seeds where available.

as boxes loosen: the paired gain on BUSI falls from 0.053 at 20 px to 0.017 at 200 px, and on CBIS-DDSM from 0.088 to not significant (per-level tests in the supplement). On the dermoscopy sets the sign flips beyond 50 px (0.033 lower at 200 px, $p < 10^{-20}$): anchoring to a base trained with tight boxes inherits its sensitivity to loose boxes in-domain. The rand100 regime buys in-domain prompt robustness at a far-OOD price that the CKA loss makes worse.

Where the anchor must be. The hook ablation (LoRA, 20 px training, tight-box evaluation, three seeds; figure in the supplement) separates the two halves of the model. With the loss on encoder activations alone, far-OOD Dice falls below the unregularised baseline in every seed, to 0.725 ± 0.023 on BUSI and 0.357 ± 0.059 on CBIS-DDSM (paired per-image loss 0.037 and 0.055, $p < 10^{-4}$), at unchanged in-domain Dice. Decoder hooks give the gains of Table 1. Anchoring both is best: 0.834 ± 0.027 and 0.626 ± 0.098 , a paired gain over decoder hooks of 0.010 and 0.088 ($p < 10^{-4}$) and over no regulariser of 0.062 and 0.175, for a 0.003 in-domain cost. The seed spread on CBIS-DDSM is large for every arm (two trainings of one configuration reached 0.536 and 0.779), so we report paired statistics rather than best seeds. The decoder is the locus, and the encoder anchor helps only once the decoder is anchored too.

A second mammography source. On DMID, which no model or probe has seen (269 images, tight boxes, table in the supplement), every comparison matches CBIS-DDSM: LoRA falls from a zero-shot 0.637 to 0.402, decoder-only, encoder-only and full fine-tuning reach 0.73, the decoder anchor lifts LoRA to 0.697 and encoder hooks alone give 0.366.

6 Results: Adaptation Budget and Drift

Figure 2 shows how the collapse depends on the budget and what moves inside the model. Decoder-only fine-tuning and encoder-only LoRA improve on both axes with every increase in budget and stay above the zero-shot far-OOD reference of 0.758 in every seed (0.840 to 0.861 and 0.811 to 0.864). LoRA with a trainable decoder is the only method that crosses it: already 0.02 to 0.05 below the others at 50 images (0.790 ± 0.027 , above zero-shot in each seed), it falls to 0.670 ± 0.031 at 250, below zero-shot in all three seeds, recovers to 0.774 ± 0.019 at 1000, above it in all three, and ends at 0.638 with all 2595 images, while

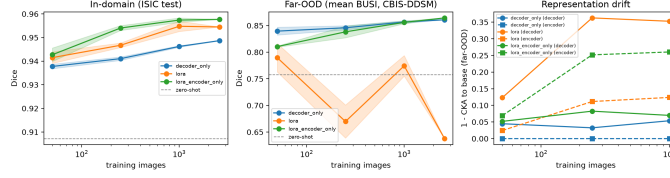


Fig. 2. Adapters trained for 6000 steps on nested subsets (log axis), zero-shot dashed. Left: in-domain Dice. Centre: far-OOD Dice (mean of BUSI, CBIS-DDSM). Right: far-OOD drift ($1 - \text{CKA}$ to the base) of the decoder’s upscaled embedding (circles) and encoder neck (squares, dashed). Dice: mean and std over three seeds at 50 to 1000 images; the 2595-image point and the drift panel are seed 0.

Table 2. AUROC for detecting per-image failures (Dice below 0.5) at tight boxes, mean $\pm$ std over three seeds (single value: seed 0 only): decoder drift (Eq. (3)) versus the decoder’s IoU estimate. Cells with fewer than 20 failures omitted; drift is zero by construction for zero-shot. AUPRC, per-seed values and the 50px table are in the supplement.

Method	BUSI (far-OOD)		CBIS-DDSM (far-OOD)		DMID (held-out)	
	drift	IoU head	drift	IoU head	drift	IoU head
Zero-shot	–	–	–	0.49	–	0.47
Decoder-only FT	–	–	–	–	0.39 ± 0.12	0.52 ± 0.09
VPT-shallow	0.95 ± 0.01	0.67 ± 0.02	0.72 ± 0.03	0.50 ± 0.06	0.57 ± 0.09	0.55 ± 0.12
VPT-deep	0.92 ± 0.01	0.64 ± 0.07	0.69 ± 0.03	0.50 ± 0.07	0.55 ± 0.09	0.50 ± 0.03
LoRA	0.96 ± 0.01	0.55 ± 0.16	0.80 ± 0.02	0.51 ± 0.09	0.69 ± 0.11	0.57 ± 0.16
Encoder-only LoRA	–	–	–	–	0.69	0.24
Full FT	–	–	–	–	0.57 ± 0.01	0.32 ± 0.03

its in-domain Dice rises monotonically. The drift panel says why. LoRA’s far-OOD decoder drift triples between 50 and 250 images, from 0.12 to 0.36, exactly where its Dice collapses, and stays there. Decoder-only fine-tuning retraining every decoder weight yet keeps far-OOD decoder drift below 0.06 at all budgets; encoder-only LoRA moves the encoder twice as much as LoRA (encoder drift 0.25 against 0.12) and keeps far-OOD decoder drift below 0.09. The collapse is thus not a symptom of too little data, of decoder training, or of encoder movement: it appears when encoder and decoder adapt jointly without an anchor. The dip at 250 images and the partial recovery at 1000 are reproduced in every seed; the recovery occurs at unchanged far-OOD decoder drift (0.35), so drift marks the onset of the collapse rather than tracking its depth.

7 Results: Label-Free Failure Detection

Drift ranks LoRA’s failures. Over three seeds LoRA fails on 61 to 82 of 647 BUSI images and 130 to 173 of 362 CBIS-DDSM images, and drift ranks these failures with AUROC 0.96 ± 0.01 and 0.80 ± 0.02 . On DMID (113 to 175 failures of 269) drift scores 0.69 ± 0.11 . The result does not hinge on the failure threshold (seed 0: AUROC 0.90 to 0.97 on BUSI and 0.73 to 0.82 on CBIS-DDSM for thresholds

from 0.3 to 0.7; supplement). Ranking quality translates into a review budget (seed 0): flagging the 20 percent of BUSI images with the highest drift captures 87 percent of LoRA’s failures and leaves a failure rate of 1.5 percent among the unflagged, from 9.4 percent overall; on CBIS-DDSM (42 percent failures) flagging half captures 79 percent (supplement). LoRA’s IoU estimate is at chance on all three sets (0.55 ± 0.16 , 0.51 ± 0.09 , 0.57 ± 0.16): after adaptation it collapses to a band between 0.4 and 0.6 regardless of mask quality.

VPT: strong on the ladder, weak on DMID. VPT-shallow and VPT-deep failures are ranked by drift with AUROC 0.95 ± 0.01 and 0.92 ± 0.01 on BUSI and 0.72 ± 0.03 and 0.69 ± 0.03 on CBIS-DDSM, but on DMID only 0.57 ± 0.09 and 0.55 ± 0.09 , with one seed of each at chance, although the per-image spread of drift there is as wide as on CBIS-DDSM. Drift is therefore not a universal signal: reliable for LoRA on the ladder, weaker and seed-dependent on DMID, weak for VPT there.

The IoU estimate reacts to the prompt, not to the modality. Full fine-tuning and encoder-only LoRA fail on only 2 to 14 ladder images at tight boxes, too few to score, and full fine-tuning’s 50 to 52 DMID failures are missed by both signals in every seed (drift 0.57 ± 0.01 , IoU estimate 0.32 ± 0.03 , below chance; encoder-only LoRA: 0.69 and 0.24). Under 50 px jitter (seed 0) their 220 to 236 CBIS-DDSM failures are detected by the IoU estimate with AUROC 0.75 to 0.79 while drift is below chance (0.23 to 0.36). The IoU head flags a poor box, which every model can sense, rather than a modality it has never seen. Drift marks predictions on which the adapted decoder has left the region the base model would use; it is uninformative once the decoder is retrained wholesale and moves for every image, or is held in place by the CKA anchor (0.35 and 0.46 on DMID). Neither signal is a self-assessment: drift needs a base-model forward, and both were validated against ground truth offline.

8 Limitations and Conclusion

All models are adapted on dermoscopy, so we have not shown that the anchors help when a far-OOD set is the training domain. The drift curves and the full-data budget point are single-seed, and far-OOD variance on mammography is large for every method. The CKA loss adds about 30 percent to training time; the drift signal needs a base-model forward and is weak on DMID; the regularised model inherits the base’s sensitivity to loose boxes in-domain.

Within those bounds, far-OOD failure of adapted MedSAM arises where the mask decoder’s representations leave the base model’s, when encoder and decoder adapt together without an anchor; a decoder anchor removes much of the failure at no in-domain cost, an encoder anchor alone makes it worse, and for LoRA the size of the departure on a single image is a label-free failure signal that holds on a domain used for nothing else.

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